Benjamin Avanzi

Professor of Actuarial Studies, The University of Melbourne

PhD (Lausanne) • FIAA • CERA • Actuary SAA • GAICD

Editor, *ASTIN Bulletin* • Associate Editor, *Insurance: Mathematics and Economics*

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Professional qualifications

FIAA	*Fellow of the Institute of Actuaries of Australia* Australian Actuaries Institute
CERA	*Chartered Enterprise Risk Actuary* via the Australian Actuaries Institute CERA Global https://ceraglobal.org/
GAICD	Graduate member of the Australian Institute of Company Directors Company Director Course (CDC), 2022
Actuary SAA	*Fully Qualified Actuary* with the Swiss Association of Actuaries (Fully recognised within the Actuarial Association of Europe)

Academic qualifications

2008 PhD	University of Lausanne, Faculty of Business and Economics (HEC Lausanne), Actuarial Science (supervisor: Prof. Hans U. Gerber)
2005 Lic Oec	University of Lausanne, Faculty of Business and Economics (HEC Lausanne), Business, major in Actuarial Science

Academic appointments

2020 -	Professor (Full) of Actuarial Studies, Centre for Actuarial Studies, Department of Economics Faculty of Business and Economics, University of Melbourne, Australia
2015 - 2020	Associate Professor of Actuarial Studies, School of Risk and Actuarial Studies UNSW Australia Business School, University of New South Wales (UNSW), Australia
2015 - 2018	Professeur associé en actuariat[1], Département de Mathématiques et de Statistique Faculté des Arts et des Sciences, Université de Montréal (UdeM), Canada
2013 - 2015	Professeur agrégé en actuariat[2], Département de Mathématiques et de Statistique Faculté des Arts et des Sciences, Université de Montréal (UdeM), Canada
2013 - 2015	Adjunct Associate Professor, School of Risk and Actuarial Studies UNSW Australia Business School[3], University of New South Wales (UNSW), Australia
2010 - 2013	Senior Lecturer of Actuarial Studies, School of Risk and Actuarial Studies Australian School of Business, University of New South Wales, Australia
2008 - 2010	Lecturer of Actuarial Studies, School of Actuarial Studies Australian School of Business, University of New South Wales, Australia
2005 - 2008	Various casual lecturer positions, University of Lausanne & Bordeaux Business School
2003 - 2008	Teaching and Research assistant, University of Lausanne

Referees

Available upon request.

[1] 'Adjunct Professor' of Actuarial Science
[2] 'Associate Professor' of Actuarial Science
[3] Formerly 'Australian School of Business'

Research

RESEARCH SNAPSHOT — 10 November 2026

5 papers 100+
papers with more than 100 Google Scholar citations each (up to 326)

26 A* | 14 A
of 46 journal articles (ABDC 2025): 87% A* or A

1,826 | h 22 | m 1.11
Google Scholar citations, h-index and m-index

$1.78m
external research funding (cash, whole-of-grant), incl. $0.57m ARC; plus lead of the $0.49m BRACE laboratory (2027–31)

15 PhD
PhD students supervised, 9 completed

7 prizes
research prizes since 2017, incl. two Hachemeister Prizes

Overview of research expertise

I am a Swiss and Australian actuary. I develop stochastic and statistical-learning models for the understanding, management, forecasting, sharing and financing of insurance and catastrophe risks. Current themes include:

- the financing of disaster resilience and adaptation: parametric insurance, climate-dependent dynamic financial analysis of insurers, and the financial resilience of households and communities after disasters (BRACE research laboratory; Natural Hazards Research Australia);
- machine learning for actuarial applications: distributional forecasting and uncertainty quantification for individual and aggregate claims reserving, high-cardinality features, and synthetic claims data;
- fairness in insurance pricing and machine learning with multiple protected attributes;
- dependence modelling of insurance claims processes, with applications to reserving, economic capital and capital allocation; and
- optimal control of surplus models: dividends, capital injections, reinsurance and risk sharing under solvency constraints.

I have also made contributions to the literature in pensions and operations management, and was Executive Chairman of a pension fund in Switzerland before becoming a full time academic. I am Editor of *ASTIN Bulletin* and Associate Editor of *Insurance: Mathematics and Economics* (both since 2018).

Major awards

2025 **Melville Actuaries Summit Prize**: awarded by the Australian Actuaries Institute for the paper *Dynamic Financial Analysis (DFA) of General Insurers under Climate Change* which was presented at the 2025 All-Actuaries Summit (Sydney, Australia)

2025 **Carol Dolan Actuaries Summit Prize**: awarded by the Australian Actuaries Institute for the paper *Dynamic Financial Analysis (DFA) of General Insurers under Climate Change* which was presented at the 2025 All-Actuaries Summit (Sydney, Australia)

2024 **Prize for Exceptional Distinction in Research**: Research Excellence (Level D-E) category, The University of Melbourne Faculty of Business and Economics and Melbourne Business School Deans' Excellence Awards.

2023 **Hachemeister Prize**, awarded by the Casualty Actuarial Society (CAS) for the paper *Ensemble distributional forecasting for insurance loss reserving* which was presented at the 2022 ASTIN Colloquium

2022 **Highly Commended Paper Prize**, awarded by United Kingdom Institute and Faculty of Actuaries for the paper *SPLICE: A Synthetic Paid Loss and Incurred Cost Experience Simulator* (published in the Annals of Actuarial Science)

2018 **Taylor Fry General Insurance Seminar Silver Prize**, awarded by an independent panel for the paper *How to proxy the unmodellable: Analysing granular insurance claims in the presence of unobservable or complex drivers* (Australian Actuaries Institute General Insurance Seminar, November 2018)

2017 **Hachemeister Prize**, awarded by the Casualty Actuarial Society (CAS) for the paper *Correlations between insurance lines of business: An illusion or a real phenomenon? Some*

methodological considerations (2016, ASTIN Bulletin, 46:2)

Fields of Research and Socio-Economic Objectives (ANZSRC 2020) – full lists in the Appendix

FoR, primary: 350206 Insurance studies; 350208 Investment and risk management; 490501 Applied statistics; 490508 Statistical data science; 490510 Stochastic analysis and modelling; 490106 Financial mathematics; 490108 Operations research. **Secondary:** 380107 Financial economics; 380202 Econometric and statistical methods; 410199 Climate change impacts and adaptation n.e.c.; 490304 Optimisation.
SEO, primary: 110201 Finance services; 110203 Superannuation and insurance services. **Secondary:** 190199 Adaptation to climate change n.e.c.; 190301 Climate change mitigation strategies; 190401/190404/190405 Climatological, hydrological and meteorological hazards; 190501 Climate change models; 190508 Impact of natural hazards caused by climate change.

Publications

Where there is an icon next to the date, then the document is published in Open Access (downloadable for free). Where available, a link to the free accepted manuscript (arXiv, Minerva Access or UNSWorks) is also given below the journal details.

Metrics

Google Scholar (10 November 2026): 1,826 citations, h-index of 22 (since 2021: 883 citations, h-index of 17)
Scopus profile (10 November 2026): 959 citations by 665 documents, 44 documents, h-index of 15
ResearchGate (10 November 2026): Research Interest Score of 831.6 (higher than 90% of all ResearchGate members, 82% of members who first published in 2005, and 95% of researchers in "Risk Management and Insurance"), 1,517 citations, h-index of 19, 14,613 reads
Web of Science Core Collection (10 November 2026): 893 citations (809 without self-citations), 45 publications, h-index of 14
m-index (Google Scholar h-index divided by years since first journal article, 2007): 1.11 (10 November 2026)

Books

2025 1. Wüthrich, Mario V., Richman, R., **Avanzi, B.**, Lindholm, M., Maggi, M., Mayer, M., Schelldorfer, J., Scognamiglio, S.
AI Tools for Actuaries
DOI: 10.2139/ssrn.5162304
Associated website: https://aitools4actuaries.com/

Submitted journal articles

2026 1. **Avanzi, B.**, Wong, B., Zhu, J.
Balancing Shareholder Value and Financial Stability under a Reduced-Form Liquidation Model
Submitted

2025 2. **Avanzi, B.**, Lambrianidis, M., Taylor, G., Wong, B.
On the use of case estimate and transactional payment data in neural networks for individual loss reserving
Resubmitted

2025 3. Lee, H. M., Antonio, K., **Avanzi, B.**, Marchi, L., Zhou, R.
How Fair, and at What Cost? Controlling Intersectional Discrimination in Predictive Models
Under revision (earlier version on arXiv as *Machine Learning with Multitype Protected Attributes*)

Peer-reviewed journal articles

Accepted manuscripts can be downloaded for free from here (2008-2019) and here (2017-) in accordance with the journals' copyright regulations.
Badges and tags: quartile according to Scimago 2025; journal rank according to the ABDC 2025 list; "FT50" means the journal is one of the 50 used in the FT research rank; "DL" means it appears on the Dean's List.

2026 4. **Avanzi, B.**, Kusch Falden, D., Steffensen, M.
When Indemnity Insurance Fails: Parametric Coverage under Binding Budget and Risk Constraints
A Q1 *Journal of Risk and Insurance*, in press, https://doi.org/10.1111/jori.70070
Accepted manuscript: arXiv:2512.21973

2026 5. **Avanzi, B.**, Li, Y., Taylor, G., Wong, B.
Dynamic Financial Analysis (DFA) of General Insurers under Climate Change
A* Q1 DL *ASTIN Bulletin*, in press, https://doi.org/10.1017/asb.2026.10102
Accepted manuscript: arXiv:2508.16444; Minerva Access
Awarded 2025 Melville Actuaries Summit Prize (Conference paper)
Awarded 2025 Carol Dolan Actuaries Summit Prize (Conference paper)

2026 6. **Avanzi, B.**, Dong, E., Laub, P., Wong, B.
Distributional Refinement Network: Distributional Forecasting via Deep Learning
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 128, 103246.
Accepted manuscript: arXiv:2406.00998; Minerva Access
YouTube summary

2025 7. **Avanzi, B.**, Tan, X., Taylor, G., Wong, B.
On the evolution of data breach reporting patterns and frequency in the United States: a cross-state analysis
A Q2 *North American Actuarial Journal*, Vol. 29, Issue 4, pp. 833–864.
Accepted manuscript: arXiv:2310.04786; Minerva Access

2024 8. **Avanzi, B.**, Li, Y., Wong, B., Xian, A.
Ensemble distributional forecasting for insurance loss reserving
A Q2 DL *Scandinavian Actuarial Journal*, Vol. 2024, Issue 9, pp. 971–1012,
Accepted manuscript: arXiv:2206.08541; Minerva Access
Awarded 2023 Hachemeister Prize.

2024 9. **Avanzi, B.**, De Felice, L.
Optimal Strategies for the Decumulation of Retirement Savings under Differing Appetites for Liquidity and Investment Risks
ABDC B Q3 *Decisions in Economics and Finance*, https://doi.org/10.1007/s10203-024-00448-y
Accepted manuscript: arXiv:2312.14355; Minerva Access

2024 10. **Avanzi, B.**, Taylor, G., Wang, M., Wong, B.
Machine learning with high-cardinality categorical features in actuarial applications
A* Q1 DL *ASTIN Bulletin*, Vol. 54, pp. 213–238.
Accepted manuscript: arXiv:2301.12710; Minerva Access

2024 11. **Avanzi, B.**, Lavender, M., Taylor, G., Wong, B.
Detection and treatment of outliers for multivariate robust loss reserving
A Q2 *Annals of Actuarial Science*, Vol. 18, pp. 102–125.
Accepted manuscript: arXiv:2203.03874; Minerva Access

2024 12. **Avanzi, B.**, Lau, H., Steffensen, M.
Optimal reinsurance design under solvency constraints
A Q2 DL *Scandinavian Actuarial Journal*, 2024(4), 383–416
Accepted manuscript: arXiv:2203.16108; Minerva Access

2023 13. **Avanzi, B.**, Lavender, M., Taylor, G., Wong, B.
On the impact of outliers in loss reserving
ABDC B Q2 *European Actuarial Journal*, Vol. 14, pp. 257–296
Accepted manuscript: arXiv:2203.00184; Minerva Access

2023 14. **Avanzi, B.**, Kusch Falden, D., Steffensen, M.
Stable Dividends under Linear-Quadratic Optimization
A Q1 *Quantitative Finance*, Vol. 23, Issue 9, pp. 1199–1215
Accepted manuscript: arXiv:2210.03494; Minerva Access

2023 15. **Avanzi, B.**, Taylor, G., Wang, M.
SPLICE: A Synthetic Paid Loss and Incurred Cost Experience Simulator
A Q2 *Annals of Actuarial Science (Actuarial Software)*, Vol. 17, Issue 1, pp. 7–35.

Accepted manuscript: arXiv:2109.04058; Minerva Access
Awarded 2022 Highly Commended Paper Prize (IFoA)

2023 16. **Avanzi, B.**, Chen, P., Henriksen, L. F. B., Wong, B.
On the surplus management of funds with assets and liabilities in presence of solvency requirements
A Q2 DL *Scandinavian Actuarial Journal*, Volume 2023, Issue 5, pp. 477-508.
Accepted manuscript: arXiv:2203.05139; UNSWorks

2022 17. Al-Mudafer, M. T., **Avanzi, B.**, Taylor, G., Wong, B.
Stochastic loss reserving with mixture density neural networks
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 105, pp. 144–174
Accepted manuscript: arXiv:2108.07924; Minerva Access

2021 18. **Avanzi, B.**, Taylor, G., Wang, M., Wong, B.
SynthETIC: an individual insurance claim simulator with feature control
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 100, pp. 296–308
Accepted manuscript: arXiv:2008.05693; Minerva Access

2021 19. **Avanzi, B.**, Lau, H., Wong, B.
On the optimality of joint periodic and extraordinary dividend strategies
A* Q1 *European Journal of Operational Research*, Vol. 295, issue 3, pp. 1189–1210
Accepted manuscript: Minerva Access

2021 20. **Avanzi, B.**, Boglioni Beaulieu, G., Lafaye de Micheaux, P., Ouimet, F., Wong, B.
A counterexample to the existence of a general central limit theorem for pairwise independent identically distributed random variables
Q1 *Journal of Mathematical Analysis and Applications*, Vol. 499, Issue 1, 124982
Accepted manuscript: arXiv:2003.01350; Minerva Access

2021 21. **Avanzi, B.**, Taylor, G., Wong, B., Yang, X.
On the modelling of multivariate counts with Cox processes and dependent shot noise intensities
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 99, pp. 9–24
Accepted manuscript: Minerva Access

2021 22. **Avanzi, B.**, Lau, H., Wong, B.
Optimal Periodic Dividend Strategies for Spectrally Negative Lévy Risk Processes With Fixed Transaction Costs
A Q2 DL *Scandinavian Actuarial Journal*, Vol. 2021, Issue 8, pp. 645–670
Accepted manuscript: arXiv:2004.01838; Minerva Access

2021 23. **Avanzi, B.**, Taylor, G., Wong, B., Xian, A.
Modelling and understanding count processes through a Markov-modulated non-homogeneous Poisson process framework
A* Q1 *European Journal of Operational Research*, Vol. 290, Issue 1, pp. 177–195
Accepted manuscript: Minerva Access

2021 24. **Avanzi, B.**, Taylor, G., Vu, P. A., Wong, B.
On Unbalanced Data and Common Shock Models in Stochastic Loss Reserving
A Q2 *Annals of Actuarial Science*, Vol. 15, Issue 1, pp. 173–203
Accepted manuscript: Minerva Access

2020 25. **Avanzi, B.**, Lau, H., Wong, B.
Optimal Periodic Dividend Strategies for Spectrally Positive Lévy Risk Processes With Fixed Transaction Costs
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 93, pp. 315–332
Accepted manuscript: Minerva Access

2020 26. **Avanzi, B.**, Taylor, G., Vu P. A., Wong, B.
A Multivariate Evolutionary Generalised Linear Model Framework with Adaptive Estimation for Claims Reserving
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 93, pp. 50–71
Accepted manuscript: Minerva Access

2018 27. **Avanzi, B.**, Taylor, G., Wong, B.
Common shock models for claim arrays

A* Q1 DL *ASTIN Bulletin*, Vol. 48, Issue 3, pp. 1109-1136
Accepted manuscript: Minerva Access

2018 28. **Avanzi, B.**, Tu, V., Wong, B.
Optimal Dividends Under Erlang(2) Inter-Dividend Decision Times
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 79, pp. 225-242
Accepted manuscript: Minerva Access

2018 29. **Avanzi, B.**, Henriksen, L. F. B., Wong, B.
On the distribution of the excedents of funds with assets and liabilities in presence of solvency and recovery requirements
A* Q1 DL *ASTIN Bulletin*, Vol. 48, Issue 2, pp. 647–672.
Accepted manuscript: UNSWorks

2017 30. **Avanzi, B.**, Pérez, J.-L., Wong, B., Yamazaki, K.
On optimal joint reflective and refractive dividend strategies in spectrally positive Lévy models
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 72, pp. 148–162.
Accepted manuscript: UNSWorks

2016 31. **Avanzi, B.**, Taylor, G., Vu, P. A., Wong, B.
Stochastic Loss Reserving with Dependence: A Flexible Multivariate Tweedie Approach
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 71, pp. 63–78.
Accepted manuscript: Minerva Access

2016 32. **Avanzi, B.**, Tu, V., Wong, B.
On the interface between optimal periodic and continuous dividend strategies in the presence of transaction costs
A* Q1 DL *ASTIN Bulletin*, Vol. 46, Issue 3, pp. 709–746.
Accepted manuscript: Minerva Access

2016 33. **Avanzi, B.**, Wong, B., Yang, X.
A micro-level claim count model with overdispersion and reporting delays
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 71, pp. 1–14.
Accepted manuscript: Minerva Access

2016 34. **Avanzi, B.**, Taylor, G., Wong, B.
Correlations between insurance lines of business: An illusion or a real phenomenon? Some methodological considerations
A* Q1 DL *ASTIN Bulletin*, Vol. 46, Issue 2, pp. 225–263.
Accepted manuscript: Minerva Access
Awarded 2017 Hachemeister Prize

2016 35. **Avanzi, B.**, Tao, J., Wong, B., Yang, X.
Capturing Non-Exchangeable Dependence in Multivariate Loss Processes with Nested Archimedean Lévy Copulas
A Q2 *Annals of Actuarial Science*, Vol. 10, Issue 01, pp. 87–117.
Accepted manuscript: Minerva Access

2016 36. **Avanzi, B.**, Tu, V., Wong, B.
A note on realistic dividends in actuarial surplus models
Q2 *Risks*, Vol. 4, Issue 4, Number 37
Accepted manuscript: Minerva Access

2014 37. **Avanzi, B.**, Tu, V., Wong, B.
On optimal periodic dividend strategies in the dual model with diffusion
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 55, pp. 210–224.
Accepted manuscript: UNSWorks

2014 38. de Treville, S., Schürhoff, N., Trigeorgis, L., **Avanzi, B.**
Optimal Sourcing and Lead-Time Reduction under Evolutionary Demand Risk
A* Q1 FT50 DL *Production and Operations Management*, Vol. 23, Issue 12, pp. 2103–2117.

2014 39. **Avanzi, B.**, Purcal, S.
Annuitisation and cross-subsidies in a two-tiered retirement saving system
A Q2 *Annals of Actuarial Science*, Vol. 8, Part 2, pp. 234–252.

Accepted manuscript: UNSWorks

2013 40. **Avanzi, B.**, Cheung, E. C. K., Wong, B., Woo, J. K.
On a periodic dividend barrier strategy in the dual model with continuous monitoring of solvency
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 52, Issue 1, pp. 98–113.

2013 41. **Avanzi, B.**, Bicer, I., de Treville, S., Trigeorgis, L.
Real Options at the Interface of Finance and Operations: Exploiting Embedded Supply Chain Real Options to Gain Competitiveness
A Q1 *European Journal of Finance*, Vol. 19, Issue 7-8, pp. 760–778.

2012 42. **Avanzi, B.**, Wong, B.
On a mean reverting dividend strategy with Brownian motion
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 51, Issue 2, pp. 229–238.

2011 43. **Avanzi, B.**, Shen, J., Wong, B.
Optimal dividends and capital injections in the dual model with diffusion
A* Q1 DL *ASTIN Bulletin*, Vol. 41, Issue 2, pp. 611–644.

2011 44. **Avanzi, B.**, Cassar, L. C., Wong, B.
Modelling Dependence in Insurance Claims Processes with Lévy Copulas
A* Q1 DL *ASTIN Bulletin*, Vol. 41, Issue 2, pp. 575–609.

2010 45. **Avanzi, B.**
What is it that makes the Swiss annuitise? A description of the Swiss retirement system
ABDC C *Australian Actuarial Journal*, Vol. 16, Issue 2, pp. 135–162.

2009 46. **Avanzi, B.**
Strategies for Dividend Distribution: A Review
A Q2 *North American Actuarial Journal*, Vol. 13, Issue 2, pp. 217–251.

2008 47. de Treville, S., Edelson, N. M., Kharkar, A. N., **Avanzi, B.**
Constructing useful theory: The case of Six Sigma
ABDC C Q1 *Operations Management Research*, Vol. 1, Issue 1, pp. 15–23.

2008 48. **Avanzi, B.**, Gerber, H. U.
Optimal dividends in the dual model with diffusion
A* Q1 DL *ASTIN Bulletin*, Vol. 38, Issue 2, pp. 653–667.

2007 49. **Avanzi, B.**, Gerber, H. U., Shiu, E. S. W.
Optimal dividends in the dual model
A* Q1 DL *Insurance: Mathematics and Economics*, Vol. 41, Issue 1, pp. 111–123.

Reports

2026 1. Jarzabkowski P., Riordan, T., Gallagher, R., Pham W., **Avanzi, B.**, Mason M., Rambaldi A., Zahnow R. , Johnson L., Wyeth S, Deng R., Taylor N., Xie Y.
Evaluation of the Resilient Homes Fund
Natural Hazards Research Australia, Report 70.2026. ISBN 978-1-923057-53-1.

2016 2. Bothwell, P. T., Kannon M. J., **Avanzi, B.**, Izzo, J. M., Knobloch, S. A., Nichols, R. S., Norris, J. L., Pan, Y., Semenovich, D., Spadola, T. A., Waite, L. M., Yarnell, D. H.
Data & Technology Working Party Report
Casualty Actuarial Society Forum, Fall 2016.

2015 3. **Avanzi, B.**, Gagné, C.
Adjustment of CIA Mortality
Report commissioned by Morneau Shepell (Montréal, Canada).

Conference publications

2018 1. **Avanzi, B.**, Taylor, G., Wong, B., Xian, A.
How to proxy the unmodellable: Analysing granular insurance claims in the presence of unobservable or complex drivers
Proceedings of the Actuaries Institute *2018 General Insurance Seminar* (peer reviewed), 12-13 November 2018 (Sydney, Australia)

2016 2. **Avanzi, B.**, Lavender, M., Taylor, G., Wong, B.
On the Impact, Detection and Treatment of Outliers in Robust Loss Reserving
Proceedings of the Actuaries Institute *2016 General Insurance Seminar* (peer reviewed), 13-15 November 2016 (Melbourne, Australia)

2010 3. **Avanzi, B.**, Wong, B.
On a mean reverting dividend strategy with Brownian motion
6th Conference in Actuarial Science & Finance on Samos (Samos, Greece)

2006 4. **Avanzi, B.**, de Treville, S.
Demand Estimation Through Credibility Theory
17th Annual Conference of POMS (Boston, U.S.A.)

2005 5. de Treville, S., Edelson, N. M., Kharkar, A. N., **Avanzi, B.**
Can we develop theory around Six Sigma? Should we care?
16th Annual Conference of POMS (Chicago, U.S.A.)

2004 6. Boisvert, C., de Treville, S., Oyon, D., **Avanzi, B.**
The Impact of Performance Measurement Systems on Lead Time Reduction
CRITOM Conference (Bocconi University, Italy)

R codes and illustrations

2024 1. **Avanzi, B.**, Ho, W., Li, Y., Wong, B., Xian, A.
CRAN package: *ADLP: Accident and Development period adjusted Linear Pools for Actuarial Stochastic Reserving*
https://CRAN.R-project.org/package=ADLP

2021 2. **Avanzi, B.**, Taylor, G., Wang, M.
CRAN package: *SPLICE: Synthetic Paid Loss and Incurred Cost Experience (SPLICE) Simulator*
https://CRAN.R-project.org/package=SPLICE

2020 3. **Avanzi, B.**, Taylor, G., Wang, M., Wong, B.
CRAN package: *SynthETIC: an individual insurance claim simulator with feature control*
https://CRAN.R-project.org/package=SynthETIC

Other R codes related to published papers, with several reproducible examples, are available from the GitHub page of Agi-Lab.

Websites

2025 1. Wüthrich, Mario V., Richman, R., **Avanzi, B.**, Lindholm, M., Maggi, M., Mayer, M., Schelldorfer, J., Scognamiglio, S.
AI Tools for Actuaries
(website associated to the book *AI Tools for Actuaries*)

2023 2. **Avanzi, B.**
Topics in Actuarial, Insurance, and Finance
(website for UniMelb ACTL20004/ACTL90021 - Sections 1.2 and 5.2 of the CM2 actuarial syllabus)

2022 3. **Avanzi, B.**
General Insurance Modelling - AMIII
(website for UniMelb ACTL30007/ACTL90020 - Sections 1 and 2 of the CS2 actuarial syllabus)

2021 4. **Avanzi, B.**
Introduction to Actuarial Studies
(website for UniMelb ACTL10001)

2021 5. **Avanzi, B.**, Boglioni Beaulieu, G., Lafaye de Micheaux, P., Xian, A., Yang, X.
Communicate data with R

Working papers

Recent working papers on arXiv (updated with accepted manuscript versions); papers since published are listed, with their accepted-manuscript links, under Peer-reviewed journal articles

Older working papers on SSRN (not updated)

2026 1. **Avanzi, B.**, Wong, B., Zhu, J.
Balancing Shareholder Value and Financial Stability under a Reduced-Form Liquidation Model
arXiv q-fin.MF

2026 2. **Avanzi, B.**, Richman, R., Wong, B., Wüthrich, M., Xie, Y.
Reinforcement Learning for Micro-Level Claims Reserving
arXiv q-fin.RM 2601.07637

2025 3. **Avanzi, B.**, Lambrianidis, M., Taylor, G., Wong, B.
On the use of case estimate and transactional payment data in neural networks for individual loss reserving
arXiv q-fin.ST 2601.05274

2025 4. Lee, H. M., Antonio, K., **Avanzi, B.**, Marchi, L., Zhou, R.
Machine Learning with Multitype Protected Attributes: Intersectional Fairness through Regularisation
arXiv cs.LG 2509.08163

2015 5. **Avanzi, B.**, Gagné, C., Tu, V.
Is Gamma Frailty a Good Model? Evidence from Canadian Pension Funds
UNSW Business School Research Paper No. 2015ACTL15

2009 6. **Avanzi, B.**, Purcal, S.
Forced savings and annuitisation with cross-subsidies: a mutation of the beast
UNSW Australian School of Business Research Paper No. 2009ACTL09

2008 7. de Treville, S., **Avanzi, B.**, Trigeorgis, L.
Rethinking Lead Time Reduction Investment: A Real Options Perspective
UNSW Australian School of Business Research Paper No. 2008ACTL16

2008 8. **Avanzi, B.**
A Review of Modern Collective Risk Theory with Dividend Strategies
UNSW Australian School of Business Research Paper No. 2008ACTL09

Research Funding

Total research funding awarded (whole-of-grant cash, as listed below): AUD 2.42 million, of which AUD 1.78 million external (including AUD 0.57 million from the ARC) and AUD 0.64 million internal (including learning and teaching grants); plus AUD 1.33 million in-kind (NHRA) and CAD 70,000 (NSERC).

National Competitive Grants

2020 - 2025 Chief Investigator, Australian Research Council (ARC) Discovery Project DP200101859
AUD 327,198 from ARC
Extreme Value Theory Approaches to Insurance in a Catastrophic Environment
Prof Qihe Tang (UNSW); A/Prof Benjamin Avanzi (University of Melbourne); A/Prof Bernard Wong (UNSW); A/Prof Jose Blanchet (Stanford)

2015 - 2020 Chief Investigator
Natural Sciences and Engineering Research Council (NSERC) of Canada, CAD 70,000
Actuarial methods for assessing diversification effects in insurance
Note: Funding for 2016–2020 (CAD 56,000) lapsed due to my return to Australia.

2013 - 2020 Lead Chief Investigator, Australian Research Council (ARC) Linkage Project LP130100723
AUD 246,892 from ARC, and AUD 88,412 from Industry Partners
Modelling claim dependencies for the general insurance industry with economic capital in view: an innovative approach with stochastic processes
Dr Benjamin Avanzi (UNSW); Dr Bernard Wong (UNSW); Dr Gregory Taylor (UNSW); Mr Stephen Britt (Insurance Australia Group Ltd); Mr Yusuf Cakan (Suncorp Group Ltd); Mr David Koob (Allianz Australia Ltd)

Other External Competitive Grants

2024 – 2027 Named researcher (both components), Natural Hazards Research Australia (NHRA) Project T4-A6a: Evaluating the Resilient Homes Fund (QLD component) and Project T4-A6b: Evaluating the Resilient Homes Program (NSW component)
AUD 1,004,825 (ex GST) cash from NHRA, and AUD 1,328,068 in-kind (The University of Queensland AUD 1,068,221; The University of Melbourne AUD 100,183; IAG, Suncorp and RACQ AUD 159,659); grant RM 2023001699
Chief Investigator: Prof Paula Jarzabkowski (UQ). Named researchers on both components: Prof Benjamin Avanzi (University of Melbourne), A/Prof Matthew Mason (UQ). QLD component: Dr Katie Meissner, Dr Laurel Johnson, A/Prof Stephanie Wyeth, Prof Alicia Rambaldi (UQ). NSW component: Dr Tyler Riordan, Dr Renee Zahnow (UQ)
Strategic partners: Queensland Reconstruction Authority (QLD component); NSW Reconstruction Authority (NSW component). Supporting industry partners (QLD component): Insurance Australia Group (IAG) – Mark Leplastrier; Suncorp – JongSok Oh; RACQ – Vanessa Fabre

2021 – 2026 Co-Partner Investigator (with Rui Zhou from University of Melbourne, and Katrien Antonio and Jan Dhaene from KU Leuven)
Global PhD Partnership KU Leuven/Melbourne GPP/21/003, 2 full PhD scholarship with travel budget, and about AUD 50,000 cash support for Partner Investigators
VALERIA: Valuation and Advanced Learning methods for Emerging, global Risks In Actuarial science

2015 – 2015 Chief Investigator (with Greg Taylor and Bernard Wong)
Australian Actuarial Research Grant, Institute of Actuaries of Australia, AUD 10,000
General forms of dependency in chain ladder structures

2012 – 2013 Chief Investigator (with Greg Taylor and Bernard Wong)
Australian Actuarial Research Grant, Institute of Actuaries of Australia, AUD 35,000
An innovative approach to stochastic claims reserving with dependence

2010 – 2011 Chief Investigator (with Bernard Wong)
Australian Actuarial Research Grant, Institute of Actuaries of Australia, AUD 20,000
On the economic value of aggregating dependent portfolios

Research Contracts

2025 – 2025 Motor Accident Insurance Commission: The use of a neural network for estimation of individual claim sizes in the presence of covariate information and real data
State of Queensland (acting through the Motor Accident Insurance Commission)
Co-investigator. Lead investigator: Professor Greg Taylor (UNSW), in collaboration with Professor Benjamin Avanzi (University of Melbourne), Dr Patrick Laub (UNSW), and Professor Bernard Wong (UNSW)

2024 – 2024 Motor Accident Insurance Commission: The use of a neural network for estimation of individual claim sizes in the presence of covariate information
State of Queensland (acting through the Motor Accident Insurance Commission)
Co-investigator. Lead investigator: Professor Greg Taylor (UNSW), in collaboration with Professor Benjamin Avanzi (University of Melbourne), Dr Patrick Laub (UNSW), and Professor Bernard Wong (UNSW)

Internal Competitive Grants

2027 – 2031 Lead Chief Investigator (co-lead: A/Prof Ben Neville), Faculty of Business and Economics Research Laboratory Fund, University of Melbourne (JUL26RLF03)
AUD 494,361 over five years (1 January 2027 – 31 December 2031), including AUD 50,000 for a symposium
BRACE (Building Resilience and Adaptation to Catastrophe Exposure): Financing Disaster Resilience and Adaptation
UoM: Prof Benjamin Avanzi, A/Prof Ben Neville, Prof Rui Zhou, A/Prof Han Li, Dr Hamza Hanbali, Mr Vincent Lau, Prof Luke Bennetts, Prof Mark Lawrence (honorary), Mr Simon O'Connor (Sustainable Finance Hub). External: Prof Paula Jarzabkowski and A/Prof Matthew Mason (UQ), Prof Mogens Steffensen (Copenhagen), Dr Debbie Kusch Falden (Liverpool); Industry Advisory Board chair: Mr Sharanjit Paddam (Finity)

2019 – 2019 Chief Investigator (with Pierre Lafaye de Micheaux)
Business School and Faculty of Science Silver Star, AUD 20,000
Complex dependence modelling to unravel relationships in Big Data

2017 – 2018 Chief Investigator (with Greg Taylor and Bernard Wong)
Business School Linkage Silver Star, AUD 40,000
Quantitative risk management and optimal control for general insurers
Partners: Insurance Australia Group Ltd; Suncorp Group Ltd

2012 – 2012 Chief Investigator
Australian School of Business Special Research Grant, AUD 9,000
Towards more realistic dividend strategies in surplus models for risk businesses

2011 – 2011 Chief Investigator (with Bernard Wong)
Australian School of Business Research Grant, AUD 25,000
Optimal dividend problems in surplus models with stochastic firm prospects

2009 – 2009 Chief Investigator (with John Evans)
UNSW 2009 International Research Collaboration Scheme, AUD 7,250
Cross-country study of social security and superannuation systems in Australia, Switzerland, and the Philippines

2009 – 2009 Chief Investigator, Australian School of Business Research Grant, AUD 18,000
Mathematical modelling of the capital of risk businesses with dividend pay-outs

Learning and Teaching Grants

2026 – 2028 Project Lead, Learning and Teaching Initiatives (LTI) Grant, The University of Melbourne, AUD 19,940
Rethinking Actuarial Education with Generative AI: What to Teach, and How

Other Funding

2016 – 2016 Network leader, School of Risk and Actuarial Studies “Insurance Risk and Capital” network, seed funding of AUD 5,000.

Supervision

PhD students supervision

2026 – Edmond Gao (UNSW with scholarship, external supervisor)
In progress

2026 – Yagebu Xie (University of Melbourne with scholarship, main supervisor)
In progress

2024 – Ho Ming Lee (University of Melbourne and KU Leuven - joint PhD with scholarship, main supervisor)
In progress

2023 – Gerry (Zherui) Li (UNSW with DP scholarship, external supervisor)
In progress

2022 – Eric (Tian) Dong (UNSW with scholarship, external supervisor)
In progress

2022 – Jim (Yanfeng) Li (UNSW with scholarship, external supervisor)
In progress

2021 – 2026 Atibhav Chaudhry
(University of Melbourne and KU Leuven - joint PhD with scholarship, co-supervisor)
Awarded

2022 – 2026 Xingyun (Claire) Tan (University of Melbourne with scholarship, co-supervisor)
Awarded

2019 – 2024 Yunshen Yang (UNSW Scientia Scholar, joint, then external joint supervisor)
Now Post-Doctoral researcher at UNSW

2017 – 2023 Guillaume Boglioni-Beaulieu

(UNSW with UIPA and Canadian scholarships, joint, then external joint supervisor)
Now Senior Lecturer at the University of New South Wales (UNSW), Sydney

2016 – 2022 Alan Xian (UNSW with APA, joint, then external supervisor)
Dean's Award for Outstanding PhD Theses
FIAA, Now Consultant at Taylor Fry Consulting, and Lecturer at Macquarie University, Sydney.

2016 – 2020 Hayden (Ka Wai) Lau (UNSW with UIPA, joint supervisor)
Now Senior Pricing Analyst at Insurance Australia Group (IAG), Sydney.

2014 – 2019 Phuong Anh Vu (UdeM and UNSW with APA - joint PhD, joint supervisor)
Awarded at UNSW, Awarded at UdeM (with "exceptional" performance)
FIAA, CERA, FCAS, Now Actuary at Munich Re America, New York.

2013 – 2017 Xinda Yang (UNSW with UIPA, joint supervisor)
Now Senior Pricing Manager at QBE Insurance, Sydney.

2013 – 2017 Vincent (Wei) Tu (UNSW with APA, joint supervisor)
FIAA, Now Manager at Taylor Fry Consulting, Sydney.

MPhil students supervision

2015 – 2016 Guillaume Boglioni-Beaulieu, with "exceptional" performance (UdeM, joint superv.)
PhD UNSW; Now Lecturer at Macquarie University, Sydney

2014 – 2015 Gwendal Combot, with "excellent" performance (UdeM, main supervisor)
Quantitative Finance (M2) graduate from École polytechnique, France.

Masters' Essay students supervision

2026 – 2027 Yanjun Jin (ACTL90012, University of Melbourne, main supervisor)
In progress

2025 – 2025 Yagebu Xie (ACTL90012, University of Melbourne, main supervisor)
Now PhD student at the University of Melbourne.

2023 – 2024 Matthew Lambrianidis (ACTL90012, University of Melbourne, main supervisor)
Now Pricing Analyst at Insurance Australia Group (IAG), Melbourne.

2020 – 2021 Xingyun (Claire) Tan (ACTL90012, University of Melbourne, main supervisor)
Now PhD student at the University of Melbourne.

Honours students supervision

2026 Gabrielle Burrell (University of Melbourne, main supervisor)
In progress

2023 Lewis de Felice, Honours Class 1, Actuarial Studies Honours Medal (University of Melbourne, main supervisor)
AIAA, Now Risk Analyst at Optiver, Sydney.

2022 Melantha (Chenyi) Wang, Honours Class 1, and the University Medal (UNSW, external supervisor)
Now Trader at VivCourt Trading, Sydney.

2021 Jim (Yanfeng) Li, Honours Class 1, and the University Medal (UNSW, external supervisor)
Now PhD student at UNSW.

2021 David Yu, Honours Class 1 (UNSW, external supervisor)
Now Graduate Trader at VivCourt Trading, Sydney.

2020 Muhammed Al-Mudafer, Honours Class 1 (UNSW, external supervisor)
FIAA; Now at nib health funds, Sydney.

2016 Mark Lavender, Honours Class 1 and the University Medal (UNSW, joint supervisor)
Now Strategic Projects Lead at Paraform (San Francisco, California).

2015 Alan Xian, Honours Class 1 and the University Medal (UNSW, joint supervisor)
PhD UNSW, FIAA
Now Manager at Taylor Fry Consulting, Sydney.

2014 Clement Yeung, Honours Class 2:1 (UNSW, external supervisor)
Now Senior Data Scientist at Unseen, Sydney.

2013 Phuong Anh Vu, Honours Class 1 (UNSW, joint supervisor)

PhD UNSW and Université de Montréal, FIAA, CERA, FCAS
Now Senior Actuary at Munich Re America, New York.

2012 Chung-Yu Liu, Honours Class 1 (UNSW, joint supervisor)
Now Data Scientist at Insurance Australia Group (IAG), Sydney.

2012 Vincent Tu, Honours Class 1 (UNSW, joint supervisor)
PhD UNSW, FIAA; Now Director at Taylor Fry Consulting, Sydney.

2011 Jamie Tao, Honours Class 1 (UNSW, joint supervisor)
Now Vice President, APAC Treasury at Jefferies, Hong Kong.

2011 Timothy Yip, Honours Class 1 (UNSW, joint supervisor)
FIAA, Now Director at Taylor Fry Consulting, Sydney.

2010 Luke Cassar, Honours Class 1 (UNSW, joint supervisor)
FIAA, CERA, Now Senior Consultant at Finity Consulting, Sydney.

2010 Jonathan Shen, Honours Class 1 and the University Medal (UNSW, joint supervisor)
FIAA, Now AI Practice Executive, Strategy & Governance, Suncorp Group, Sydney.

Referee and review activities

Editorial duties

2018 - *ASTIN Bulletin*, Editor

2018 - *Insurance: Mathematics and Economics*, Associate Editor

2018 - 2018 *ASTIN Bulletin*, Member of the Editorial Board

2016 - 2023 *Risks*, Member of the Editorial Board

Journals refereed (Web of Science records)

Operations Research
European Journal of Operational Research
Insurance: Mathematics and Economics
ASTIN Bulletin
Scandinavian Actuarial Journal
North American Actuarial Journal
Journal of Risk and Insurance
Annals of Actuarial Science
The Geneva Papers on Risk and Insurance: Issues and Practice
Asia-Pacific Journal of Risk and Insurance
European Actuarial Journal
Variance (CAS Publications)
Risks
Mathematical Finance
Statistics and Probability Letters
Journal of Applied Probability / Advances in Applied Probability
Journal of Computational and Applied Mathematics
Journal of Optimization Theory and Applications
SIAM Journal on Control and Optimization
Stochastic Models
Japanese Journal of Statistics and Data Science
Economic Modelling
Expert Systems with Applications
Communications in Statistics - Theory and Methods
Applied Stochastic Models in Business and Industry
Filomat
Applied Mathematical Modelling
Mathematical Problems in Engineering
Mathematical Methods in the Applied Sciences
Journal of Inequalities and Applications

IEEE Transactions on Pattern Analysis and Machine Intelligence
International Journal of Information Technology & Decision Making
Intelligent Systems in Accounting, Finance and Management
Discrete Dynamics in Nature and Society
IMA Journal of Management Mathematics
Science China Mathematics
Applied Mathematics - A Journal of Chinese Universities
Bulletin of the Malaysian Mathematical Sciences Society

National Competitive Research Grant agencies

Australia: Australian Research Council (ARC): DP, LP, DECRA, NISDRG, FL, IC, FT
Canada: National Science and Engineering Research Council (NSERC)
Switzerland: National Science Foundation (SNSF)
Poland: National Science Centre (NSC)
Hong Kong Research Grants Council (RGC)

External Promotions

UNSW (Sydney, Australia): 2019 (twice), 2024, 2025
Macquarie University (Sydney, Australia): 2020
University of Chinese Academy of Sciences (Beijing, China): 2022

Scientific committee of conferences

2026	ASTIN Bulletin Conference at ETHZ (Zurich, Switzerland)
2020 –	Member of the Advisory Board, One World Actuarial Research Seminar
2020 –	Conference in Actuarial Science & Finance on Samos (Samos, Greece)
2020	Online International Conference in Actuarial science, data science and finance
2018 *[Chair]*	22nd International Congress on Insurance: Mathematics and Economics (Sydney, Australia)
2013	New York Annual Conference of Asia-Pacific Risk and Insurance Association (APRIA)
2012	Seoul Annual Conference of Asia-Pacific Risk and Insurance Association (APRIA)
2011	Tokyo Annual Conference of Asia-Pacific Risk and Insurance Association (APRIA)
2010	Singapore World Risk and Insurance Economics Congress (WRIEC)
2009	Beijing Annual Conference of Asia-Pacific Risk and Insurance Association (APRIA)

Organising committee of conferences

2019	4th Workshop on Risk: modelling, optimization and inference (with applications in Finance, Insurance and Superannuation) (Sydney, Australia)
2017	3rd Workshop on Risk: modelling, optimization and inference (with applications in Finance, Insurance and Superannuation) (Sydney, Australia)

Organising of short events

2022	Uncertainty in decision-making: game-theoretic and machine learning approaches Speakers: Professor Jose Blanchet (Stanford), Professor Henry Lam (Columbia) University of Melbourne, 9 December 2022

Miscellaneous reviewing activities

2012 Skipper Award Committee (APRIA)
2009 Skipper Award Committee (APRIA)

PhD jury

2024 Dec. ARANDJELOVIĆ Aleksandar, Theory of Neural Networks with Applications in Finance, Insurance and Climate-Economy Modelling, Macquarie University. External examiner.

2021 Jan. ARAUJO ACUNA, José Carlos, Tempering and seasonality in non-life insurance modeling, University of Lausanne. External examiner.

2016 Sep. JIN, Can (Ken), Occupation time problem in some insurance risk models, University of Melbourne. External examiner.

2014 Dec. OMIDI-FIROUZI, Hassan, On the design of customized risk measures in insurance, the problem of capital allocation and the theory of fluctuations for Lévy processes, Université de Montréal. Chair.

2014 Oct. RAMLI, Siti Norafidah Mohd, Modelling Multivariate Dependence Structures in Insurance and Credit Risk via Copulas, Macquarie University. External examiner.

2014 Aug. SIMARD, Clarence, Modélisation du carnet d'ordres limites et prévision de séries temporelles, Université de Montréal. Chair.

2014 Mar. FARD Farzad Alavi, Analysis of Pricing Financial Derivatives Under Regime-Switching Economy, Macquarie University. External examiner.

Academic visits

2026 Sep. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2024 Jul. UQ Business School, University of Queensland, Brisbane (Australia)

2024 Feb. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2023 Sep. Department of Mathematical Science, University of Copenhagen (Denmark)

2023 Sep. Bayes Business School, London (United Kingdom)

2022 Dec. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2022 Nov. Department of Mathematical Science, University of Copenhagen (Denmark)

2022 Feb. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2021 Jun. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2020 Feb. School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2019 Aug. Department of Mathematical Science, University of Copenhagen (Denmark)

2019 Jul. Institut de Science Financière et d'Assurances, Université Lyon 1 (France)

2019 Jun.-Aug. Department of Actuarial Science, University of Lausanne (Switzerland)

2019 Mar. Department of Mathematical Science, University of Copenhagen (Denmark)

2019 Jan.-Feb. Institut de Science Financière et d'Assurances, Université Lyon 1 (France)

2018 Feb. Department of Analytics & Operations, National University of Singapore (Singapore)

2015 Jan. School of Risk and Actuarial Studies, University of New South Wales (Australia)

2014 July Department of Statistics and Actuarial Science, University of Hong Kong

2014 Mar. School of Risk and Actuarial Studies, University of New South Wales (Australia)

2014 Mar. Centre for Actuarial Studies, University of Melbourne (Australia)

2012 Dec. Department of Mathematics and Statistics, Université de Montréal (Canada)

2012 June Department of Statistics and Actuarial Science, University of Hong Kong

2012 Jan. School of Economics and Political Science, University of St. Gallen (Switzerland)

2012 Jan. Department of Operations, University of Lausanne (Switzerland)

2010 Nov. Department of Insurance and Risk Management, The Wharton School, UPENN (U.S.A.)

2010 Nov. Institute of Insurance Economics, University of St. Gallen (Switzerland)

2010 June Department of Management, University of Lausanne (Switzerland)

2009 Dec. School of Actuarial Studies, Université Laval (Québec City, Canada)

2009 Dec. Department of Mathematics and Statistics, Université de Montréal (Canada)

2009 Dec. Department of Statistics and Actuarial Science, University of Waterloo (Canada)

2009 June Department of Actuarial Science, University of Lausanne (Switzerland)

Presentations

Conference Presentations (invited speaker)

2026 Mar. *Machine Learning for Claims Reserving: From Flexible Distributional Models to Reinforcement Learning*
Plenary Speaker at AI in risk assessment and mitigation Workshop, ICMS, Bayes Centre, Edinburgh, 4–6 March 2026 (Edinburgh, UK)

2019 Sep. *Stochastic dependence and general insurance capital: from data to models*
Plenary Speaker at 6th Workshop on "Recent developments in dependence modelling with applications in finance and insurance" (Agistri, Greece)

2018 Nov. *Modelling insurance claim counts and reporting delays with Cox processes*
Research School of Finance, Actuarial Studies and Statistics – 2018 Summer Camp. (Bowral, Australia)

2017 Dec. *Modelling insurance claim counts and reporting delays with Cox processes*
UNSW-MacquarieU Workshop on Risk: Modelling, optimization and inference (Sydney, Australia)

2017 Nov. *Correlations between lines of business - Are they real or illusionary?*
CAS 2017 Annual Meeting (Anaheim, CA, USA)

2014 July *On optimal periodic dividend strategies in the dual model with diffusion*
Gerber-Shiu Workshop, University of Hong Kong (Hong Kong)

2011 Nov. *Modelling Dependence in Insurance Claims Processes with Lévy Copulas*
Advances in Financial Risk Workshop, Macquarie University (Sydney, Australia)

2010 Dec. *Claim dependence modelling with Lévy copulas*
International Conference in Applied Statistics and Financial Mathematics (Hong Kong)

Conference Presentations (peer-reviewed submissions)

2026 May *Beyond Pairwise Correlation: Capturing Nonlinear and Higher-Order Dependence with Distance Measures*
2026 All Actuaries Summit (Melbourne, Australia)
(joint with Ho Ming Lee, Bernard Wong, Guillaume Boglioni Beaulieu, Pierre Lafaye de Micheaux, Rui Zhou)

2026 Jan *When Indemnity Insurance Fails: Parametric Coverage under Binding Budget and Risk Constraints*
1st ASTIN Bulletin Conference at ETH Zurich (Switzerland)
(joint with Debbie Kusch Falden and Mogens Steffensen)

2025 Jun *Uncertainty Quantification in Neural Network Models for Actuarial Use*
2025 All Actuaries Summit (Sydney, Australia)
(joint with Tian (Eric) Dong (presenter), Patrick Laub, and Bernard Wong)

2025 Jun *Including Climate Change in Dynamic Financial Analysis: Long-term Outlook for the General Insurance Market*
2025 All Actuaries Summit (Sydney, Australia)
(joint with Yanfeng (Jim) Li (presenter), Greg Taylor, and Bernard Wong)
Awarded the Melville All Actuaries Prize, and the Carol Dolan All Actuaries Prize

2023 Sep *Fairness through regularization: an approach to mitigate group disparities for multiple protected features*
8th Workshop on "Recent Developments in Dependence Modelling with Applications in Finance, Insurance and Pensions" (Agistri, Greece)

2023 May *Spoilt for Choice: Dealing with Many Categorical Features in Actuarial Machine Learning*
2023 International Congress of Actuaries
(joint with Greg Taylor, Melantha Wang, and Bernard Wong)

2022 July *Stochastic Ensemble Loss Reserving*
25th International Congress on Insurance: Mathematics and Economics (Online)
(joint with Yanfeng Li, Bernard Wong, and Alan Xian)

2022 July — *Optimal reinsurance under terminal value constraints*
25th International Congress on Insurance: Mathematics and Economics (Online)
(joint with Hayden Lau and Mogens Steffensen)

2022 June — *Stochastic Ensemble Loss Reserving*
2022 Virtual ASTIN/AFIR Colloquium, June 20-24, 2022
(joint with Yanfeng Li, Bernard Wong, and Alan Xian)

2022 June — *Optimal reinsurance under terminal value constraints*
2022 Virtual ASTIN/AFIR Colloquium, June 20-24, 2022
(joint with Hayden Lau and Mogens Steffensen)

2022 May — *Harnessing the power of "The Wisdom of Crowds": How do we optimize the ensembling of different loss reserving models?*
2022 All-Actuaries Summit: Getting closer, Thinking bigger (Australia)
(joint with Yanfeng Li, Bernard Wong, and Alan Xian)

2022 May — *An AI-driven Approach to Quantifying Model Error in Loss Reserving Generalised Linear Models*
2022 All-Actuaries Summit: Getting closer, Thinking bigger (Australia)
(joint with Greg Taylor, Bernard Wong, and David Yu)

2022 May — *SPLICE: Synthetic data generation for loss experience*
2022 All-Actuaries Summit: Getting closer, Thinking bigger (Australia)
(joint with Greg Taylor, Melantha Wang, and Bernard Wong)

2021 May — *Neural Networks in Reserving: how and why are they worth considering?*
2021 All-Actuaries Virtual Summit: Thriving in an Age of Extremes (Australia)
(joint with M. Al-Mudafer, Greg Taylor, and Bernard Wong)

2019 Nov. — *When (not) to use abstract dependence structures: theoretical and practical considerations*
10th Australasian Actuarial Education and Research Symposium (Melbourne, Australia)

2019 July — *Inference of counts using Markov-modulated non-homogeneous Poisson processes*
23rd International Congress on Insurance: Mathematics and Economics (Munich, Germany)

2019 Apr — *A tractable method for unravelling and modelling unobservable or complex dependence drivers (with granular data)*
IAA Section Colloquium 2019 (Cape Town, South Africa)

2018 June — *Hybrid strategies in the presence of fixed transaction costs*
10th Conference in Actuarial Science & Finance on Samos (Samos, Greece)

2017 July — *Optimal Dividends Under Erlang(2) Inter-Dividend Decision Times*
21st International Congress on Insurance: Mathematics and Economics (Vienna, Austria))

2016 Sep — *On the distribution of the excedents of funds with assets and liabilities in presence of solvency and recovery requirements*
3rd European Actuarial Journal Conference (Lyon, France)

2016 May — *On the interface between optimal periodic and continuous dividend strategies in the presence of transaction costs*
9th Conference in Actuarial Science & Finance on Samos (Samos, Greece)

2015 Dec — *Correlations between insurance lines of business: An illusion or a real phenomenon? Some methodological considerations*
7th Australasian Actuarial Education and Research Symposium (Gold Coast, Australia)

2014 July — *Claim dependencies in economic capital modeling: The Australian experience*
49th Actuarial Research Conference (Santa Barbara, USA)

2013 July — *On optimal periodic dividend strategies in the dual model with diffusion*
48th Actuarial Research Conference (Philadelphia, USA)

2012 Dec. — *On a mean reverting dividend strategy with Brownian motion*
5th Australasian Actuarial Education and Research Symposium (Melbourne, Australia)

2012 July — *Optimal dividends in a perturbed Cramér-Lundberg model with stochastic prospects and a penalty at ruin*
4th International Gerber-Shiu Workshop (Melbourne, Australia)

2012 June — *Optimal dividends in a perturbed Cramér-Lundberg model with stochastic prospects and a penalty*

at ruin
16th International Congress on Insurance: Mathematics and Economics (Hong Kong)

2011 Nov. *What is it that makes the Swiss annuitise? A description of the Swiss retirement system*
2011 Australasian Actuarial Education and Research Symposium (Canberra, Australia)

2011 June *Modelling Dependence in Insurance Claims Processes with Lévy Copulas*
15th International Congress on Insurance: Mathematics and Economics (Trieste, Italy)

2011 June *Optimal dividends and capital injections in the dual model with diffusion*
15th International Congress on Insurance: Mathematics and Economics (Trieste, Italy)

2011 June *Modelling Dependence in Insurance Claims Processes with Lévy Copulas*
Memorable Actuarial Research Conference (Leuven, Belgium)

2010 June *On a mean reverting dividend strategy with Brownian motion*
6th Conference in Actuarial Science & Finance on Samos (Samos, Greece)

2009 Sep. *Forced savings and annuitisation with cross-subsidies: a mutation of the beast*
AFIR Colloquium (Munich, Germany)

2009 Sep. *What is it that makes the Swiss annuitise? A description of the Swiss retirement system*
LIFE Colloquium (Munich, Germany)

2009 May *On the level of national retirement savings with annuitisation and cross-subsidies: a two-tiered economic model*
13th International Congress on Insurance: Mathematics and Economics (Istanbul, Turkey)

2009 May *What is it that makes the Swiss annuitise? A description of the Swiss retirement system*
13th International Congress on Insurance: Mathematics and Economics (Istanbul, Turkey)

2008 Oct. *On Optimal Dividend Strategies: Review and Dual Model*
UNSW Actuarial Studies Research Symposium 2008 (Sydney, Australia)

2008 Jul. *A Review of Modern Collective Risk Theory with Dividend Strategies*
12th International Congress on Insurance: Mathematics and Economics (Dalian, China)

2007 Jul. *Optimal dividends in the dual model with diffusion*
11th International Congress on Insurance: Mathematics and Economics (Piraeus, Greece)

2006 Jul. *Optimal dividends in the dual model*
10th International Congress on Insurance: Mathematics and Economics (Leuven, Belgium)

2006 May *Demand Estimation Through Credibility Theory*
17th Annual Conference of POMS (Boston, USA)

Academic Seminars

2023 Sep *Fairness through regularization: an approach to mitigate group disparities for multiple protected features*
Bayes Business School Seminars (London, UK)

2022 Dec *On the mitigation of unwanted discrimination of multiple protected features via distance covariance regularisation*
2022 UNSW Workshop on Risk and Actuarial Frontiers (Sydney, Australia)

2022 Jun *Ensemble distributional forecasting for insurance loss reserving*
One World Actuarial Research Seminar

2022 Mar *Optimal reinsurance under terminal value constraints*
Department of Econometrics and Business Statistics, Monash Business School, Melbourne (Australia)

2021 Nov *Optimal reinsurance under terminal value constraints*
School of Risk and Actuarial Studies, UNSW Sydney (Australia)

2019 Jul *A Multivariate Evolutionary Generalised Linear Model Framework with Adaptive Estimation for Claims Reserving*
Séminaires L^2 Lyon-Lausanne (ISFA, Lyon)

2019 Jun *Stochastic dependence and general insurance capital: from data to models*
Centre for Actuarial Studies, Department of Economics, University of Melbourne (Australia)

2019 Mar *Inference and prediction of counts using Markov-modulated non-homogeneous Poisson processes*

Department of Mathematical Science, University of Copenhagen (Denmark)

2019 Feb — *Inference and prediction of counts using Markov-modulated non-homogeneous Poisson processes*
Séminaire Labo, Institut de Science Financière et d'Assurances, Université Lyon 1 (France)

2016 Aug — *On the distribution of the profits of assets-liability funds in presence of regulation*
Macquarie University Actuarial Studies Seminar, Macquarie University (Sydney, Australia)

2015 Dec — *Correlations between insurance lines of business: An illusion or a real phenomenon? Some methodological considerations*
School of Risk & Actuarial Studies Research Day 2015 (Sydney, Australia)

2015 Apr. — *Dépendance et assurance: une histoire de corrélation?*
Club Mathématique de l'Université de Montréal, Montreal (Canada)

2015 Jan. — *Actuarial applications of Lévy copulas*
Actuarial and Financial Mathematics Seminar, Quantact Lab, Center for Research in Mathematics (CRM), Montreal (Canada)

2014 Mar. — *On optimal periodic dividend strategies in the dual model with diffusion*
Centre for Actuarial Studies, University of Melbourne (Australia)

2012 Dec. — *Dividendes et injections de capital dans le modèle dual : Stratégies optimales*
Séminaires du Centre de Recherche en Mathématiques, Université de Montréal (Canada)

2010 Nov. — *An Analysis of Private Pensions in Switzerland, A Country Where People Choose to Save and then Annuitize*
Rosen-Huebner-McCahan Seminar, The Wharton School, UPENN (Philadelphia, U.S.A.)

2010 Nov. — *Forced savings and annuitisation: an international perspective*
University of St. Gallen (Switzerland)

2010 Apr. — *On a mean reverting dividend strategy with Brownian motion*
UNSW Actuarial Studies Research Seminar (Sydney, Australia)

2009 Dec. — *On a mean reverting dividend strategy with Brownian motion*
Séminaires du Centre de Recherche en Mathématiques, Université de Montréal (Canada)

2009 Dec. — *On a mean reverting dividend strategy with Brownian motion*
Université Laval (Québec, Canada)

2009 Dec. — *Forced savings and annuitisation with cross-subsidies: a mutation of the beast*
Department of Statistics and Actuarial Science Seminars, University of Waterloo (Canada)

2009 Dec. — *On a mean reverting dividend strategy with Brownian motion*
Department of Statistics and Actuarial Science Seminars, University of Waterloo (Canada)

2009 Mar. — *The demand for retirement annuities and non-mandatory superannuation savings: the case of Switzerland*
Macquarie University Actuarial Studies Seminar, Macquarie University (Sydney, Australia)

2009 Jan. — *The demand for retirement annuities and non-mandatory superannuation savings: the case of Switzerland*
International Research Workshop (UNSW, Sydney, Australia)

2008 Feb. — *Optimal dividends in the dual model with diffusion*
UNSW Actuarial Studies Research Seminar (Sydney, Australia)

2006 Dec. — *Optimal Dividends in the Dual Model*
Séminaires L^2 Lyon-Lausanne (ISFA, Lyon)

Other attended conferences

2024 May — 12th Conference of Actuarial Science and Finance (Samos, Greece)

2022 Nov. — 2022 Australasian Actuarial Education and Research Symposium (Canberra, Australia)

2019 Mar. — Workshop "Fair Valuation in Insurance" (Brussels, Belgium)

2016 Jun. — CAS Course on Professionalism (Toronto, Canada)

2015 Jun. — Enterprise Risk Management Symposium (Washington, USA)

2015 Jun. — The Changing Face of GI, Actuaries Institute (Sydney, Australia)

2014 July — 16th International Congress on Insurance: Mathematics and Economics (Shanghai, China)

2010 Nov. OECD/IOPS Global Forum on Private Pensions, on invitation (Sydney, Australia)

2010 Sep. 6th International Longevity Risk and Capital Markets Solutions Conference (Sydney, Australia)

2010 Jul. 18th Australian Colloquium of Superannuation Researchers, *Retirement Saving Choices: challenges for individuals, industry and public policy* (Sydney, Australia)

2009 Jul. 17th Australian Colloquium of Superannuation Researchers, *Celebrating 100 years of a National Retirement Income System - The Challenges Ahead* (Sydney, Australia)

2009 Jun. 2b) or not 2b) Conference (Lausanne, Switzerland)

2008 Nov. 1st Australasian Actuarial Education and Research Symposium (Sydney, Australia)

Teaching

Awards

2016 2015 School of Risk and Actuarial Studies Teaching Innovation Award, for "Use of Learning Catalytics in the Actuarial Program"

2011 Nominated for a 2012 Vice-Chancellor Award for Teaching Excellence

2011 Australian School of Business Teaching Excellence Award, for "Outstanding Technology-Enabled Teaching Innovation"

Teaching Innovations using Technology

2026 – 2028 *Rethinking Actuarial Education with Generative AI: What to Teach, and How*, University of Melbourne Learning and Teaching Initiatives grant (project lead, AUD 19,940; see Research Funding, Learning and Teaching Grants). Examines how actuarial curriculum and delivery should adapt to generative AI, and in particular what students must still learn to do themselves and what can be left to AI, through a literature review and consultation with the profession. Designs, deploys and evaluates a grounded GenAI study companion in ACTL20005/ACTL90002 (from Semester 2, 2027, about 150 students a year), comparing foundational with applied content and undergraduate with postgraduate cohorts. Outputs: a position paper, an empirical SoTL study, and a reusable how-to guide and template for other quantitative subjects. Builds on the 2026 Cogniti pilot below.

2026 S2 ACTL20004 and ACTL90021 Cogniti pilot, part of the University of Melbourne's initiative on generative-AI teaching agents. Designed and deployed course-specific agents grounded in the subject materials: an *Exercise Coach* that gets students unstuck through graduated hints without revealing solutions (preserving the learning in the attempt), and a *Readiness Diagnostic* that checks assumed prerequisite knowledge and returns a prioritised revision plan. The principle-based design is reusable across subjects, and is paired with website guidance on using students' own LLMs responsibly.

2024 S1 ACTL20004 and ACTL90021 digital exam pilot, part of the University of Melbourne's ongoing initiative to enhance assessment methods through digital solutions. On-campus, open book, invigilated exam with fleet devices provided so that students can answer questions using Excel. Main objectives were to support digitisation of assessment, while making assessment more authentic.

2024 S1 ACTL30007 and ACTL90020 digital exam pilot, part of the University of Melbourne's ongoing initiative to enhance assessment methods through digital solutions. On-campus, open book, invigilated exam with fleet devices provided so that students can answer questions using R. Main objectives were to support digitisation of assessment, while making assessment more authentic.

2022 - 2024 Production and online publication of lecture slides using `RMarkdown`, then `Quarto`, to facilitate sharing of R (and python) code with students.

2021 - 2024 Randomisation of final exam questions to minimise risk of cheating in online exams, using R and its package `exams`, to automatically produce randomised data analysis and quiz questions

banks to use on Canvas. Use was developed from randomisation of quiz answers (simplest) to fully randomised questions and answers, and finally fully individualized data sets for each students.

2021 S2 Yellowdig pilot, ACTL10001 Introduction to Actuarial Studies

2017 Bluepulse pilot at UNSW (rolled out in 2019 university wide)

2017 First to use Teams for large class teaching at UNSW wide (and one of the first worldwide – Teams was still a beta software)

2016 Initiated the development of the "Lightboard" for recording lectures (top half of presenter can be seen through the board, where they write) - UNSW wide

2015 "Learning catalytics" (Harvard active engagement tool) pilot at UNSW Business

2011 Use of Grademark for online assignments (UNSW Business)

2010 Introduction of rubrics in actuarial assignments (UNSW)

2009 - 2010 Full flipping of ACTL2001 Financial Mathematics with the recording of >60 videos, and the introduction of "lectorials"

2009 Use of Turnitin for online assignments, including R code (UNSW Business)

Education

2017 'Presenting online' (one day), Australia's National Institute of Dramatic Art (NIDA)

2017 'Public Speaking Bootcamp' (one day), Australia's National Institute of Dramatic Art (NIDA)

2009 FULT 'Foundations of University Learning and Teaching' (five days), University of New South Wales

Courses taught

2027 Sep ISS-2027 Summer School *Actuarial Data: From Statistics to Decisions*, Swiss Association of Actuaries at the University of Lausanne (lecturer, with Martin Bladt)

2026 S2 ACTL 90021 Topics in Insurance and Finance (M.Sc., University of Melbourne)

2026 S2 ACTL 20004 Topics in Actuarial Studies (B.Sc., University of Melbourne)

2025 S2 "Disaster Risk Finance" in the Executive training course on Disaster Management and Disaster Resilient Infrastructure (Exec Ed, University of Melbourne)

2024 S2 ACTL 90021 Topics in Insurance and Finance (M.Sc., University of Melbourne)

2024 S2 ACTL 20004 Topics in Actuarial Studies (B.Sc., University of Melbourne)

2024 S1 ACTL 90020 General Insurance Modelling (M.Sc., University of Melbourne)

2024 S1 ACTL 30007 Actuarial Modelling III (B.Sc., University of Melbourne)

2023 S2 ACTL 90021 Topics in Insurance and Finance (M.Sc., University of Melbourne)

2023 S2 ACTL 20004 Topics in Actuarial Studies (B.Sc., University of Melbourne)

2023 S1 ACTL 90020 General Insurance Modelling (M.Sc., University of Melbourne)

2023 S1 ACTL 30007 Actuarial Modelling III (B.Sc., University of Melbourne)

2022 S1 ACTL 90020 General Insurance Modelling (M.Sc., University of Melbourne)

2022 S1 ACTL 30007 Actuarial Modelling III (B.Sc., University of Melbourne)

2021 S2 ACTL 10001 Introduction to Actuarial Studies (B.Sc., University of Melbourne)

2021 S1 ACTL 90020 General Insurance Modelling (M.Sc., University of Melbourne)

2021 S1 ACTL 30007 Actuarial Modelling III (B.Sc., University of Melbourne)

2020 S2 ACTL 10001 Introduction to Actuarial Studies (B.Sc., University of Melbourne)

2020 S1 ACTL 90020 General Insurance Modelling (M.Sc., University of Melbourne)

2020 S1 ACTL 30007 Actuarial Modelling III (B.Sc., University of Melbourne)

2019 T6 ZZBU 6507 Data Visualisation & Communication – developed but not delivered (fully online, Master of Analytics, UNSW)

2019 T3 ACTL 1101 Introduction to Actuarial Studies (B.Sc., UNSW)

2019 Jul Actuarial Journals and the Publishing Process (PhD, University of Lyon 1)

2018 S2 ACTL 1101 Introduction to Actuarial Studies (B.Sc., UNSW)

2018 S1	ACTL 4003/5003 Research Methods in Actuarial Science (Honours and HDR, UNSW)
2017 S2	ACTL 1101 Introduction to Actuarial Studies (B.Sc., UNSW)
2017 S2	ACTL 4003/5003 Research Methods in Actuarial Science (Honours and HDR, UNSW)
2016 S2	ACTL 1101 Introduction to Actuarial Studies (B.Sc., UNSW)
2016 S1	ACTL 3141 Actuarial Statistics (B.Sc., UNSW)
2015 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2015 S2	ACTL 3162 Insurance Risk Models (B.Sc., UNSW)
2014 Fall	ACT3284 Modèles en assurances IARD (B.Sc., Université de Montréal)
2014 Fall	ACT3241 Théorie du Risque (B.Sc., Université de Montréal)
2013 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2013 S2	ACTL 3003 Insurance Risk Models (B.Sc., UNSW)
2013 S1	ACTL 5104 Actuarial Statistics (M.Sc., UNSW)
2013 S1	ACTL 3001 Actuarial Statistics (B.Sc., UNSW)
2012 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2012 S2	ACTL 3003 Insurance Risk Models (B.Sc., UNSW)
2011 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2011 S2	ACTL 3003 Insurance Risk Models (B.Sc., UNSW)
2011 S1	ACTL 5102 Financial Mathematics (M.Sc., UNSW)
2011 S1	ACTL 2001 Financial Mathematics (B.Sc., UNSW)
2010 SS	ACTL 5102 Financial Mathematics (M.Sc., UNSW)
2010 SS	ACTL 2001 Financial Mathematics (B.Sc., UNSW)
2010 S1	ACTL 5102 Financial Mathematics (M.Sc., UNSW)
2010 S1	ACTL 2001 Financial Mathematics (B.Sc., UNSW)
2009 S2	ACTL 4003/5003 Research Topics in Actuarial Science (Honours and HDR, UNSW)
2009 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2009 S1	ACTL 5102 Financial Mathematics (M.Sc., UNSW)
2009 S1	ACTL 2001 Financial Mathematics (B.Sc., UNSW)
2008 S2	ACTL 5106 Insurance Risk Models (M.Sc., UNSW)
2008 S1	Gestion des opérations (B.Sc., University of Lausanne)
2007 S2	Credibility Theory (M.Sc., University of Lausanne)
2007 S1	Gestion des opérations (B.Sc., University of Lausanne)
2005 S2	Laboratory of selected topics in actuarial mathematics (M.Sc., University of Lausanne)
2005 S1	Mathematics of Lead Time (MAS, Bordeaux Business School)

Masters project supervision (ACTL5004)

2016	Ka Wai (Hayden) Lau (UNSW) Now pricing specialist at IAG.

Summer Research Scholarships

2019-2020	Melantha (Chenyi) Wang (UNSW)
2019-2020	Simon (Wanzhang) Jing (UNSW)
2018-2019	Wilson Cheng (UNSW)
2016-2017	Guanting Liu (UNSW)
2016-2017	Edward Djizmedjian (UNSW)

Internal Engagement and Leadership

Awards

2018 **The Dean's Distinguished Leadership Award**, in recognition of exceptional and sustained contributions to the achievement of the goals of the UNSW School of Business and Strategy 2025.
This award may be given to an individual member of staff who, over a sustained period, has provided meritorious and distinguished leadership that has significantly exceeded the scope and commitment of usual expectations in achieving Business School goals.

Academic engagement and leadership

2026 - 2027 TALQAC/PROVOST Program review panel member, Doctor of Clinical Dentistry

2026 - 2027 2026 Higher Degrees by Research Review panel member, HDR degrees of the Faculty of Science

2021 - 2024 Course Academic Progress Committee (CAPC) meetings
– Chair (12/2024) – Member (07/2021, 01/2022, 01/2023)

2023 - 2024 TALQAC/PROVOST Program review panel chair, Master, Graduate Diploma, Graduate Certificate in Urban Horticulture, Master of Ecosystem Management and Conservation, Graduate Certificate in Bushfire Planning and Management, Graduate Certificate in Arboriculture, Graduate Certificate in Garden Design

2023 - 2023 Panel member, Undergraduate student misconduct hearing

2022 - 2023 Member, *Working group on diversifying evidence of teaching effectiveness*

2021 - 2022 Elected Member, Teaching and Learning Quality Assurance Committee (TALQAC), The University of Melbourne
– TALQAC/PROVOST Program review panel member, Master of Contemporary Art, Graduate Certificate in Visual Art, Master of Film and Television, Master of Production and Master of Screenwriting (2022)
– TALQAC/PROVOST Program review panel member, Bachelor of Design (2021)

2020 - Member, Academic Board, The University of Melbourne
– Appeals committee member (2020: 1)

2019 - 2019 Elected member, Faculty Board, UNSW Business School, UNSW
(representing the Faculty Academic Staff)

2019 - 2019 Member, School of Risk and Actuarial Studies Research Committee, UNSW

2017 - 2018 Chair, UNSW Business School 3+ Reference Group

2017 - 2018 Member, UNSW 3+ Transition Advisory Group

2017 - 2018 Member, UNSW 3+ Engagement Reference Group

2017 - 2018 Elected member, Faculty Board, UNSW Business School, UNSW
(representing the School of Risk and Actuarial Studies)

2016 - 2017 Chair, Alternative Delivery Modes Committee, School of Risk and Actuarial Studies, UNSW

2015 - 2017 Associate Head of School, School of Risk and Actuarial Studies, UNSW

2015 - 2017 Chair, Learning & Teaching Committee, School of Risk and Actuarial Studies, UNSW

2015 - 2017 Member, Faculty Education Committee, UNSW Business School, UNSW

2015 - 2017 Member, School Industry Advisory Board, School of Risk and Actuarial Studies, UNSW

2017 - 2017 Member, Education Focussed roles selection committee, UNSW Business School

2015 - 2016 Member, Faculty Board, UNSW Business School, UNSW

2014 - 2015 Chair, website committee, Departement of Mathematics and Statistics, UdeM

2014 - 2015 Member, 'internationalisation' committee, Departement of Mathematics and Statistics, UdeM

2012 - 2013 Elected chair of the *School Plenum*, School of Risk and Actuarial Studies, UNSW

2012 - 2013 Elected member of the *School Advisory Board*, School of Risk and Actuarial Studies, UNSW

2012 - 2013 Postgraduate Research Coordinator, School of Risk and Actuarial Studies, UNSW

2012 - 2013 Honours Coordinator, School of Risk and Actuarial Studies, UNSW

2012 – 2013 Member, *Australian School of Business Promotions Committee for Promotion to Senior Lecturer*, UNSW

2009 – 2012 Academic representative, *Australian School of Business E-learning Committee*, UNSW

2009 – 2012 Director, Actuarial Studies Co-op Program of the UNSW

2011 – 2011 Member, *ASB Linkage Grant Working Party*, UNSW

2010 – 2010 Member, *Australian School of Business Standing Committee*, UNSW

2007 – 2008 Elected member, Commission of the University of Lausanne Personnel (official legal representative of all (administrative and academic) staff employed by the university)

2006 – 2008 Elected member, Faculty Council, Faculty of Business and Economics of the University of Lausanne

2006 – 2008 Treasurer, *Association du Corps Intermédiaire et des Doctorants de l'Université de Lausanne* (official representative of the regular (teaching and research) staff that is below the professor level)

2006 – 2008 Member, Committee *Enseignement* (teaching) of the University of Lausanne

2006 – 2008 Member, Committee *Recherche* (research) of the University of Lausanne

Teaching leadership

Seminars and presentations

2026 (10 November) *AI and AI agents to enhance student learning: What we built in ACTL20004/90021, what happened, and what comes next*, 3rd Annual Melbourne EconTALE Symposium, Department of Economics (University of Melbourne)

2022 (2 February) Yellowdig presenter, *WCLA PD: Student Engagement Pilot Projects of 2021 (Piazza & YellowDig)*

2020 (14 September) Panelist, *WCLA Symposium: Adapting to Radical Disruptions in Business Education*

2018 (February) *Digital Uplift* interview, Business Digital Learning (UNSW Business School, UNSW).

2017 (October, 18) *ACTL1101 Digital Uplift*, School of Risk and Actuarial Studies Digital Uplift Planning (UNSW Business School, UNSW).

2016 (February, 25) *Assessment Good Practice*, School of Risk and Actuarial Studies Learning & Teaching Workshop (UNSW Business School, UNSW).

2015 (November, 3) *Spotlight on Program Learning Goals in the School of Risk and Actuarial*, UNSW Business School – Academic's eLearning Showcase & Workshop (UNSW Business School, UNSW).

2014 (December, 5) Seminar on the flipped classroom to the FOLIUM team (teaching support) of the University of Montreal as well as some flipped classroom pilot project leaders, University of Montréal, Canada

2014 (October, 1) Seminar on how and when to use technology (in particular, tablet PCs and recorded videos) for teaching technical subjects, made to interested members of the Department of Mathematics and Statistics, University of Montréal, Canada

2014 (September, 8) Brief presentation on how and when to use technology (in particular, tablet PCs and recorded videos) for teaching technical subjects, made to the Department Assembly, Department of Mathematics and Statistics, University of Montréal, Canada

2014 (May, 23) Presentation on how and when to use technology (in particular, tablet PCs and recorded videos) for teaching technical subjects, made to the steering committee of the project *Mathéma-TIC*, Montréal, Canada

2013 (August, 16) Member, *Large Class Teaching Panel*, Fundamentals of University Learning and Teaching (FULT), Learning & Teaching @ UNSW

2012 (June, 21) Member, *Large Class Teaching Panel*, Fundamentals of University Learning and Teaching (FULT), Learning & Teaching @ UNSW

2012 (April, 3) *Low-tech high-impact teaching strategies for large classes*, Connections in Learning and Teaching (Learning & Teaching @ UNSW)

2011 (November, 8) *Using videos in teaching: when and how?*, ASB Academics' Showcase of Technology in Teaching (ASB, UNSW)

2011 (February, 17) Member, *Large Class Teaching Panel*, Fundamentals of University Learning and Teaching (FULT), Learning & Teaching @ UNSW

2010 (March, 1) *Australian School of Business Strategic Retreat*, invited for the eLearning component

2010 (August, 26) Member, *Large Class Teaching Panel*, Fundamentals of University Learning and Teaching (FULT), Learning & Teaching @ UNSW

2009 (November, 19) Member, *Large Class Teaching Panel*, Fundamentals of University Learning and Teaching (FULT), Learning & Teaching @ UNSW

2009 (November, 18) Recorded interview by Colina Mason (Learning & Teaching @ UNSW) for the sessional staff gateway resource

2009 (August, 28) School of Accounting seminar (UNSW), *Digital Tablets and Screen Recording*

2009 (July, 8) ASB Workshop (UNSW), *Digital Tablets and Screen Recording*

Other

2016 S1 *FULT peer observation* participant: interested FULT participants can come and observe in my lectures, as well as access the course website.

Other

Research leadership

2027 – Lead (with A/Prof Ben Neville), BRACE research laboratory (Building Resilience and Adaptation to Catastrophe Exposure), Faculty of Business and Economics, The University of Melbourne

2025 – 2025 Panellist, "The Diplomacy of Engaging Senior Policymakers and CEOs: A Forum for Senior Researchers", FBE Researcher Development Program (RDP) Forum for Senior Researchers

Formal Mentorship/Coaching

2020 – University of Melbourne academic advising program
(2024: 20 meetings, 2023: 47 meetings, 2022: 21 meetings, 2021: 29 meetings, 2020: 1 meeting)

2022 – 2025 PhD student at the University of Bayreuth, CoMento Program

2020 – 2020 University of Melbourne level C

2019 – 2019 UNSW Business School PhD Fellow

2017 – 2017 1 UNSW Scientia Scholar

2016 – 2016 1 UNSW Scientia Scholar

2015 – 2016 1 ECR academic from UNSW Business School

ECR development (external)

2023 – 2023 Guest lecture on how to be a good reviewer (from an Editor's perspective)
MQBS Meet the Academic Seminar Series - Session 5: How to become a good reviewer...
Macquarie Business School PhD Students, Sydney, Australia (via Zoom)

2022 – 2022 Guest lecture on Research assessment, Actuarial research, and the Publication process
ISFA PhD students, Université Bernard, Lyon, France (via Zoom)

2019 – 2019 Guest lecture on Research assessment, Actuarial research, and the Publication process
ISFA PhD students, Université Bernard, Lyon, France (in person)

External Engagement and Leadership

Professional appointments (selected)

2023 – 2028	Non-Executive Director, Management Board of the Theatre Royal (Hobart, Tasmania) Member, then Chair of the Finance, Audit, and Risk Committee (FRAC)
2021 – 2022	Elected Councillor, Caulfield Junior College Convenor, Policy Review and Development Subcommittee
2020 – 2022	Elected Board Member, École Française Melbourne (EFM) Vice-President since 2021
2018 – 2019	Non-Executive Director, Lycee Condorcet the International French School of Sydney
2014 – 2015	Actuarial consulting, Service de consultation en actuariat (Université de Montréal)
2008 – 2012	Non-Executive Director, *Fondation Comoedia*, Switzerland (group insurance)
2007 – 2008	Founder, Executive Director, *Fondation Comoedia*, Switzerland (group insurance)
2006 – 2008	Executive Director, Chair, *Fondation de prévoyance Artes et Comoedia*, Switzerland (pension fund, 200 employers, 2000 insured)
2004 – 2004	Consultant, *La Suisse Assurances* (actuarial department for group life), Lausanne, Switzerland

Professional affiliations

2018 –	Member, Australian Institute of Company Directors (AICD)
2014 –	Academic Member, Casualty Actuarial Society (CAS)
2008 –	Member (Fellow since 2023), Actuaries Institute of Australia[4]
2007 –	Member, Swiss Association of Actuaries (full member of IAA)
2009 –	Member, Actuarial Studies in Non-Life Insurance section of IAA (ASTIN)
2008 –	Member, American Risk and Insurance Association (ARIA)
2008 – 2011	Member, Asia-Pacific Risk and Insurance Association (APRIA)

Professional service

2026 –	External reviewer, CERA Global Association (CGA) Review Panel (representative from the Australian Actuaries Institute)
2014 –	Member, CAS Academic Central
2009 – 2016	Member, Hachemeister and Michelbacher prizes committee (CAS)
2014 – 2016	Member, Data and Technology Working Party (CAS)

In the media

Press

2026
1. Interview with Laura Mayer (ABC Ballarat) about car insurance premiums and rise of theft, aired on radio by Steve Martin on *Ballarat Breakfast* 24 March 2026, 7:50am (go to 1h40 through the program).

2024
2. Liu, Q., *As international student numbers rise, calls for urgent reform on pregnancy health cover* The Citizen, 31 January 2024.

Professional websites and blogs

2021
1. Chen, J., *Virtual Summit Shorts: Neural networks in reserving – how and why are they worth considering?* Actuaries Digital, 9 June 2021. (Sydney, Australia)

[4] Inactive in 2013 and 2014.

BusinessThink

2025 1. *Data breach reporting crisis: How delays impact cyber insurance risks*
15 October 2025

2019 2. *A smarter approach to analysis refines insurance predictions*
11 October 2019

2015 3. *Are insurance companies holding too much capital?*
16 December 2015; republished in UNSW Newsroom, 18 January 2016

Other non academic publications

2026 1. **Avanzi, B.**
Unaffordable and unavailable: the case for parametric home insurance
Actuaries Digital, The Magazine of the Actuaries Institute, 16 October 2026. (Sydney, Australia).

2024 2. **Avanzi, B.**, De Felice, L.
Towards Optimising the Decumulation of Retirement Savings
Actuaries Digital, The Magazine of the Actuaries Institute, November 2024. (Sydney, Australia).

2020 3. **Avanzi, B.**, Taylor, G., Wang, M., Wong, B.
SynthETIC: A simulator of Synthetic Experience Tracking Insurance Claims
Kasa AI Blog, 14 September 2020.

2019 4. **Avanzi, B.**, Taylor, G., Wong, B., Xian, A.
How to proxy the unmodellable
Actuaries Digital, The Magazine of the Actuaries Institute, April 2019. (Sydney, Australia).

2016 5. **Avanzi, B.**, Taylor, G., Wong, B.
Construction of detailed correlation structures across GI business segments
Actuaries Digital, The Magazine of the Actuaries Institute, 14 December 2016. (Sydney, Australia).

2015 6. **Avanzi, B.**, Taylor, G., Wong, B.
Are correlations real or imagined?
Actuaries Digital, The Magazine of the Actuaries Institute, 30 September 2015. (Sydney, Australia).

2014 7. **Avanzi, B.**, Taylor, G., Wong, B.
Research into claim dependencies: an industry and academic collaboration
Actuaries, The Magazine of the Actuaries Institute, August issue, pp. 9–11. (Sydney, Australia).

Industry presentations

2025 Nov. Dynamic Financial Analysis (DFA) of General Insurers under Climate Change
Plenary session, 2025 IAG Actuarial Summit

2025 Oct. Dynamic Financial Analysis (DFA) of General Insurers under Climate Change
ClimateScoop, Finity Consulting (Sydney, Australia)

2025 Oct. Dynamic Financial Analysis (DFA) of General Insurers under Climate Change: Overview and Potential Applications to AASB S2 disclosures
Insurance Council of Australia (ICA) Net Zero Working Group (Sydney, Australia)

2017 Sep. *Recent outlier detection methods with illustrations in loss reserving*
'Insights' session of the Actuaries Institute (Sydney, Australia)

2016 Nov. *On the Impact, Detection and Treatment of Outliers in Robust Loss Reserving*
Actuaries Institute 'General Insurance Seminar' (Melbourne, Australia)

2016 Oct. *Construction of detailed correlation structures across GI business segments*
'Insights' session of the Actuaries Institute (Sydney, Australia)

2016 Sep. Non-life stream presentation of the 'Insurance Risk and Capital' network workshop on *Recent advances and open problems in Insurance Risk and Capital regulation*

2016 Apr. *Diversification in Insurance: A Capital Question*

Learn@Lunch (Sydney, Australia)

2016 Apr. *Beyond Correlation: Recent Developments in Modelling Claims Dependencies*
'Insights' session of the Actuaries Institute (Sydney, Australia)

2015 Oct. *Correlations between insurance lines of business: Illusion or phenomenon?*
'Insights' session of the Actuaries Institute (Sydney, Australia)

Other presentations

2021 Oct. Invited Speaker, 2021 Contact-less Night by Actuarial Students' Society

Events

2016 Organiser, 'Insurance Risk and Capital' network workshop on *Recent advances and open problems in Insurance Risk and Capital regulation*, with Professor Shaun Wang (NTU Singapore) and Robert Thomson (Head of Actuarial services, Australian Prudential Regulation Authority).

Miscellaneous written contributions

2019 Jan. *Welcome to new students*
Finity Actuarial Bulletin

2018 Jan. *Welcome to new students*
Finity Actuarial Bulletin

2016 Jan. *Welcome to new students*
Finity Actuarial Bulletin no 77

Appendix: ANZSRC 2020 codes

Fields of Research[5]

Primary:

- 3502 Banking, finance and investment
  - 350206 Insurance studies
  - 350208 Investment and risk management
- 4905 Statistics
  - 490501 Applied statistics
  - 490508 Statistical data science
  - 490510 Stochastic analysis and modelling
- 4901 Applied mathematics
  - 490106 Financial mathematics
  - 490108 Operations research

Secondary:

- 3801 Applied economics
  - 380107 Financial economics
- 3802 Econometrics
  - 380202 Econometric and statistical methods
- 4101 Climate change impacts and adaptation
  - 410199 Climate change impacts and adaptation not elsewhere classified
- 4903 Numerical and computational mathematics
  - 490304 Optimisation

Socio-Economic Codes[6]

Primary:

- 1102 Financial services
  - 110201 Finance services
  - 110203 Superannuation and insurance services

Secondary:

- 1901 Adaptation to climate change
  - 190199 Adaptation to climate change not elsewhere classified
- 1903 Mitigation of climate change
  - 190301 Climate change mitigation strategies
- 1904 Natural hazards
  - 190401 Climatological hazards (e.g. extreme temperatures, drought and wildfires)
  - 190404 Hydrological hazards (e.g. avalanches and floods)
  - 190405 Meteorological hazards (e.g. cyclones and storms)
- 1905 Understanding climate change
  - 190501 Climate change models
  - 190508 Understanding the impact of natural hazards caused by climate change

[5] See, e.g., https://www.abs.gov.au/statistics/classifications/australian-and-new-zealand-standard-research-classification-anzsrc/2020/anzsrc2020_for.xlsx.

[6] See, e.g., https://www.abs.gov.au/statistics/classifications/australian-and-new-zealand-standard-research-classification-anzsrc/2020/anzsrc2020_seo.xlsx.